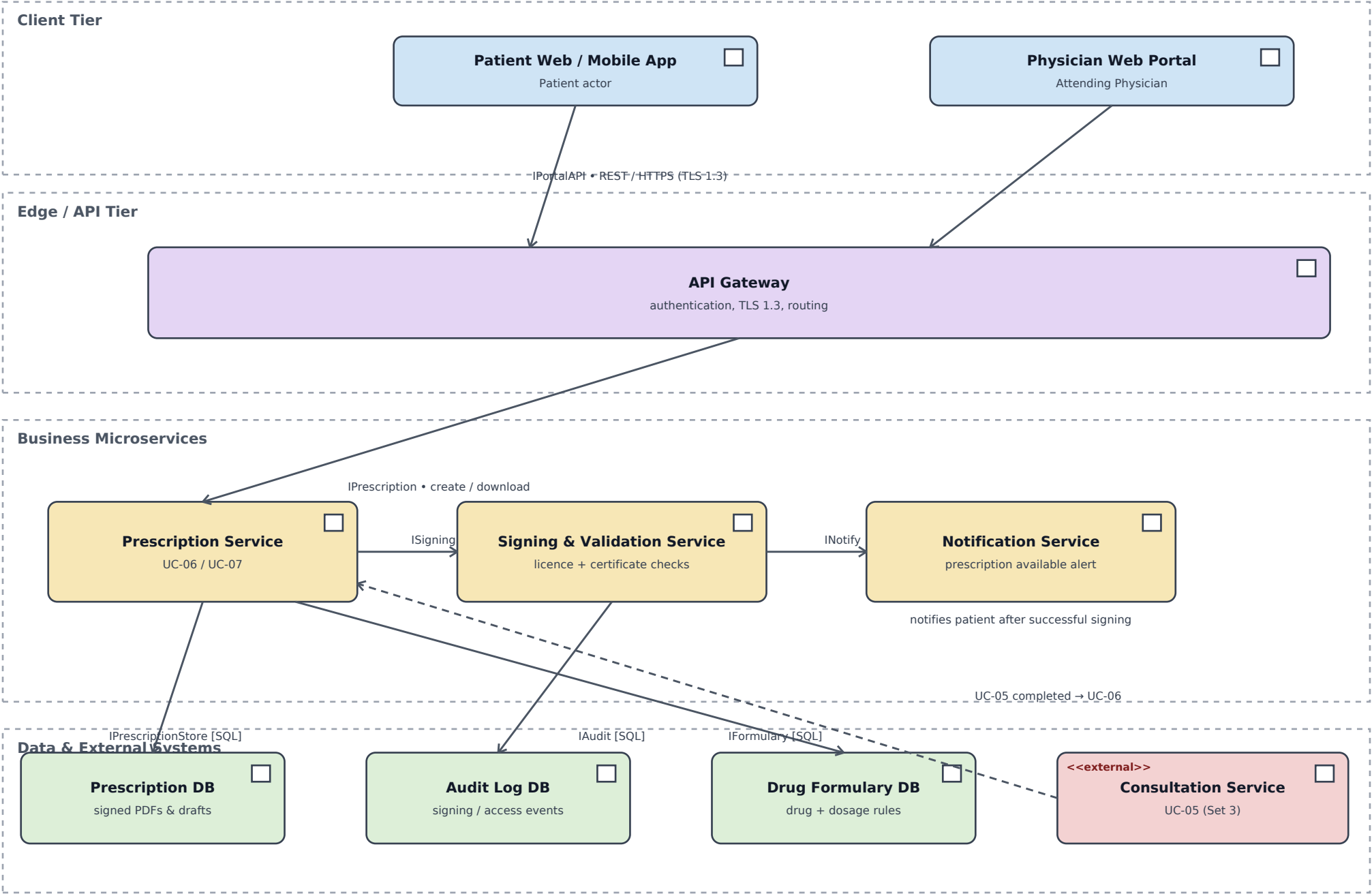


Telemedicine Slot Booking & Prescription Portal - Microservices Architecture

UML Component Diagram - Set 4: Prescription Lifecycle (UC-06 & UC-07)



Legend —●— service/data dependency - - - - cross-set dependency

Set 4 owns UC-06 & UC-07 • UC-06 extends UC-05 from Set 3

Lab 3: Architecture Selection Justification

Telemedicine Slot Booking & Prescription Portal — Set 4: Prescription Lifecycle

Architectural Choice: Microservices Architecture

Applied to Set 4: UC-06 Author & Digitally Sign Prescription and UC-07 Download Prescription PDF.

We chose Microservices Architecture for the Prescription Lifecycle because the requirements separate prescription authoring, digital-signature validation, storage, audit logging and patient notification into clear responsibilities. The design keeps these responsibilities independently deployable while the API Gateway provides a controlled entry point.

1. Independent scaling and focused performance

Prescription creation and prescription PDF download have different workloads. The Prescription Service can be scaled independently from the Notification and Signing/Validation services. This is useful because FR-005 requires the signed PDF to remain available for patient download within 24 hours, while FR-001 requires reliable generation and digital signing of the prescription.

2. Security and controlled access

FR-001 requires a valid cryptographic signature and valid physician licence information. A dedicated Signing & Validation Service keeps licence and signing-certificate checks separate from ordinary prescription processing. The API Gateway authenticates users and terminates TLS 1.3, while the Prescription DB and Audit Log DB are accessed through their owning services. This limits unnecessary exposure of sensitive patient records.

3. Fault isolation and auditability

The requirements state that when signing fails, no prescription is issued; the draft is retained, the failure reason is recorded in the audit log, and the patient is not notified. Separating the signing, prescription, audit and notification responsibilities makes this failure path easier to isolate and retry without losing the draft.

4. Replaceability and maintainability

Notification delivery can change without changing the core prescription logic. Likewise, signing or validation implementation can evolve independently as long as the service interface remains stable. This reduces the impact of changes to one part of the Prescription Lifecycle.

5. Fit with the existing portal architecture

Set 4 is connected to Set 3 because UC-06 extends UC-05, Conduct Video Consultation. The proposed microservice design therefore keeps the cross-set dependency explicit: the Prescription Service starts from a completed consultation, then creates and signs the prescription and makes it available for UC-07 download.

Styles considered

Layered Architecture would be simpler, but tightly coupled prescription, signing and notification logic would make independent deployment and fault isolation harder. Client-Server would centralize the processing and increase the impact of a server-side failure. The accepted trade-off of Microservices is higher operational complexity and the need to manage service communication carefully.